Where to start?

The abrasive itself, sand paper if you like, has to be the first thing to look at. Your abrasive is a tool, just like your gouges, so buy a good quality tool. Cheap abrasive is, like so many things, false economy.

The abrasive I use is known as Rhynogrip, which is an aluminium oxide abrasive with a velcro type backing, so it can be used for hand sanding or on an arbor for power sanding and has all of the required attributes.

While it is a premium product, it actually isn't as expensive as you might think. There are of course other abrasives around, but I believe this is one of, if not the best, on the market.

Rhynogrip abrasive

GOOD QUALITY ABRASIVE SHOULD BE:

- Fast cutting and efficient without producing unnecessary heat
- Flexible and able to roll and fold into the shapes required by turners
- Long lasting
- Available in a good range of grits
- Versatile, so no matter what you make, you can use the same abrasive: who wants to have different sets of abrasive for different jobs?

Keep it fresh

I have heard a few turners tell me that they save old abrasive and use it as fine finishing paper, suggesting that old 240 grit becomes 320 grit after use. It doesn't: old 240 grit abrasive is just that, old and worn out. Throw it away and cut a new bit. The effect that these turners are experiencing is actually burnishing, rather than sanding. Fresh abrasive will cut, or more correctly scratch, the surface of the wood: each consecutive grit leaving progressively finer and finer scratches until they are invisible to the naked eye. Old abrasive doesn't cut or scratch, it burnishes, which is the same effect that traditional pole-lathe turners would use, picking up a handful of shavings and rubbing them on the work to leave a slightly shiny surface. The definition of burnish is 'to polish a surface by friction', so you will make the surface more shiny by burnishing it with old abrasive, but you won't remove tool marks or the scratches of the previous grit. So if you want to burnish, then go ahead, but sand properly first! Once your abrasive looks worn or at the first sign of it not working properly, having used every part of it, throw it out and cut some more.

Pressure

The amount of pressure that you apply to the work can have an effect on the quality of finish. The problem is that it is virtually impossible for me to quantify how much pressure you should be applying to the work. Not enough and sanding will take an age and still not achieve the desired results.

Too much pressure and you can deform the work and produce too much heat, which is never a good thing for the abrasive or, more importantly, the wood. You do need to apply pressure, but if hand sanding, you shouldn't hurt or burn your fingers. If power sanding, the drill motor shouldn't sound like it's straining; it should run normally.

So if you are sanding but don't feel like you are getting the result you should, try applying a little more pressure. If you are sanding and burn your fingers or get through drills on a regular basis, you might want to ease back a little.

TOP TIP: AVOID HEAT

Heat is a bad thing when sanding. It shortens the life of your abrasive and can permanently damage the wood, so should be avoided wherever possible. Starting to sand at too high a grit level, will mean that you need to sand for longer than you should. By starting at a lower grit, the sanding will be easier and therefore quicker, meaning that you will produce less heat. A trick I picked up somewhere along the line, is to rotate the abrasive when hand sanding, so that it, and your fingers, never get too hot against the wood.

Some woods will handle heat better than others, but some species including yew (*Taxus baccata*), cherry (*Prunus* spp.) and many exotics will develop heat cracks in their end grain under excessive sanding. These tiny little cracks cannot be sanded away and can run surprisingly deep; they need to be removed with a sharp tool.

Lathe speed

Generally speaking, the advice for gauging turning speed is to run as fast as is safely possible, without vibration or the lathe walking across the workshop floor. For sanding, the speed needs reducing. The exact speed you use will depend on what is available to you on your lathe, but as a guide, if you hand sand at about half the speed that you are turning, and power sand at 500rpm or below, you should be about right. If the lathe is running too fast it doesn't give the abrasive a chance to work and produces unnecessary heat, which will shorten the life of your abrasive and potentially spoil the work.

Hand vs power sanding

Power sanding uses a drill and arbor with a velcro type backing to hold the abrasive

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